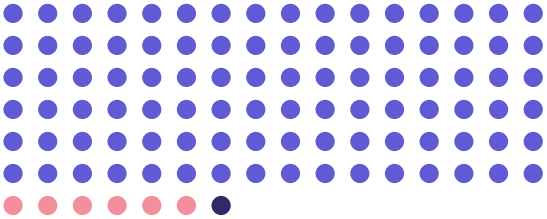
PUBLIC UNDERSTANDING / SURVEY FINDINGS

Who responded, and what stood out?

Emerging biomedical technologies and Alzheimer's disease treatments

103 valid responses / 11 questions / descriptive results

Q1 / AGE PROFILE



93.2%

96 respondents aged 18-30

6 under 18; 1 aged 46-60. All other age bands: 0.
Each dot represents one respondent.

Q2 / RELEVANT EDUCATION

Currently studying / working in a related field	35.9% (37)
Previously received relevant education	18.4% (19)
No relevant educational background	37.9% (39)
Not sure	7.8% (8)

Q3 / COGNITIVE-DISORDER EXPERIENCE

Direct experience, including caregiving	4.9% (5)
Some exposure; no long-term caregiving	23.3% (24)
No relevant experience	69.9% (72)
Prefer not to answer	1.9% (2)

THREE PRIORITIES TO CARRY INTO MATERIAL DESIGN

68.9%
Actual improvement
Prioritized in Q8

49.5%
Unspecified improvement
A barrier selected in Q10

51.5%
Illustrations / flowcharts
Preferred format in Q11

Q1-Q3; highlights from Q8, Q10 and Q11 | n=103 | Source: ORCA aggregate survey results.

FAMILIARITY / INTERPRETATION OF EVIDENCE

How familiar? How convincing?

Self-rated familiarity and statement responses are displayed separately.

Q4 / RNA-EDITING FAMILIARITY



Never heard	Term only	Basic concepts	Can explain	Quite familiar
10.7%	23.3%	30.1%	29.1%	6.8%

Q5 / DOES EACH STATEMENT MATCH YOUR UNDERSTANDING?

Mostly matches Mostly does not match Not sure

01 RNA can be edited without directly rewriting genomic DNA.



n=67 / n=15 / n=21

02 Methods may differ in duration, safety and maturity.



n=76 / n=11 / n=16

03 Mechanism alone is insufficient to judge patient suitability.



n=74 / n=11 / n=18

04 Animal-measure improvement usually means patient efficacy.



n=29 / n=53 / n=21

51.5% **selected "mostly does not match" for the animal-to-patient statement.**

53 respondents; 29 (28.2%) selected "mostly matches" and 21 (20.4%) were unsure.

Q4-Q5 | n=103 | Source: ORCA aggregate survey results.

Multiple-choice percentages use all 103 respondents; options can overlap and totals can exceed 100%.

RESEARCH VIGNETTES / Q6 & Q7

Reading early research

Both vignettes are hypothetical survey scenarios, not real treatment results.

Q6 / MULTIPLE CHOICE

40%

Average biomarker decrease

Possible biological effect; symptoms and long-term outcomes still need assessment

69.9% (72)

Measure changed; clinical value remains uncertain

42.7% (44)

Fairly clear clinical benefit already demonstrated

30.1% (31)

Worth using; suitability mainly judged by a physician

29.1% (30)

Not sure

9.7% (10)

Q7 / SINGLE CHOICE

Early human study

Small sample; some biological changes; further follow-up needed

Research value; clinical benefit still needs verification

39.8% (41)

Human biological changes preliminarily prove therapeutic value

14.6% (15)

Prioritize further improvement in biological measures

17.5% (18)

Further use mainly depends on patient acceptance

5.8% (6)

Need more information before forming a judgment

22.3% (23)

A COMMUNICATION TASK

Biological change

What changed in a study measure?

→

Patient benefit

What improved for the patient?

→

Remaining uncertainty

What still needs follow-up?

Q6-Q7 | n=103 | Source: ORCA aggregate survey results.

Multiple-choice percentages use all 103 respondents; options can overlap and totals can exceed 100%.

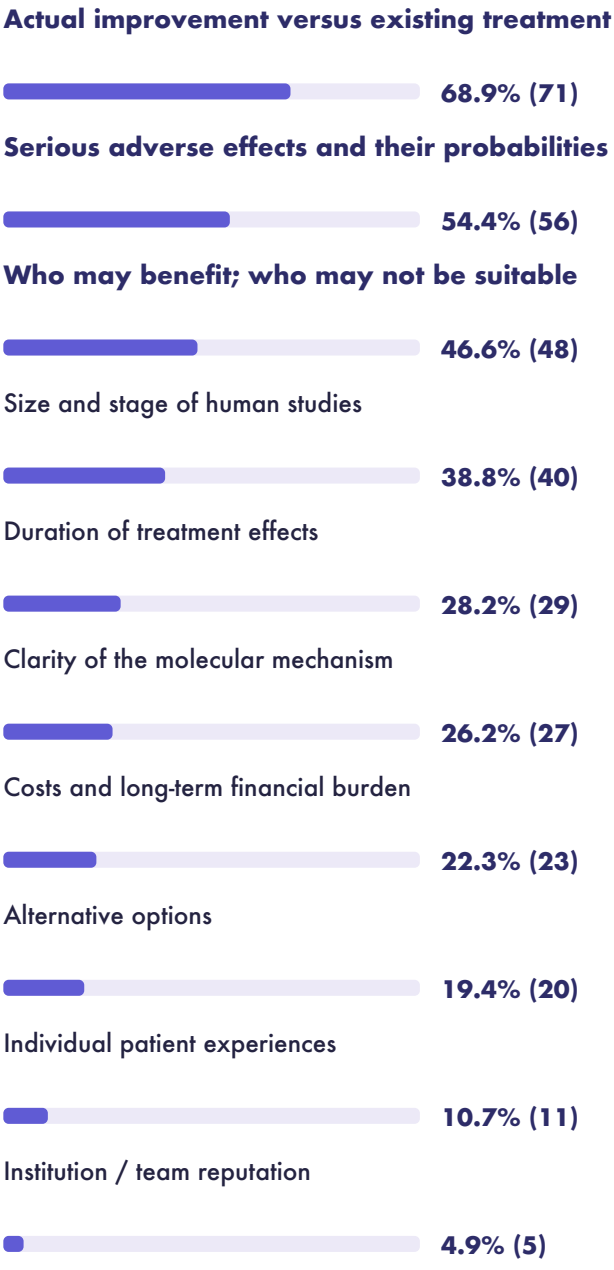
DECISION NEEDS / COMMUNICATION BARRIERS

What to explain first - and what to fix

Treatment priorities and comprehension barriers appear side by side.

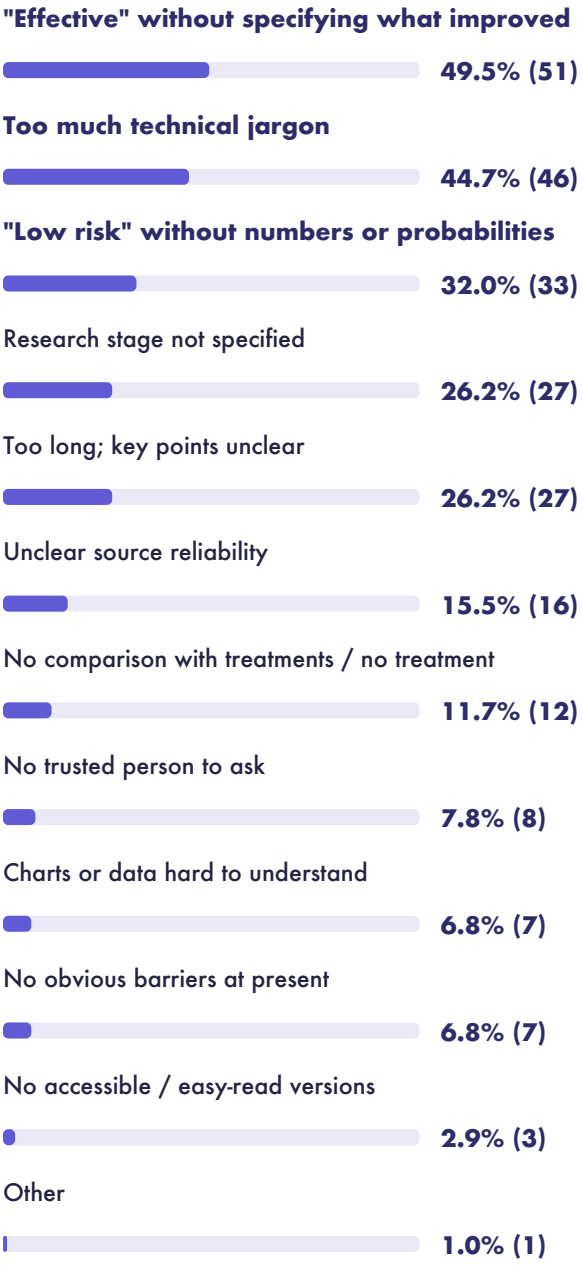
Q8 / INFORMATION PRIORITIES

Select up to four



Q10 / BARRIERS TO UNDERSTANDING

Select up to three



Design implication: specify the actual improvement, translate technical terms, and quantify serious risks.

Q8 and Q10 | n=103 | Source: ORCA aggregate survey results.

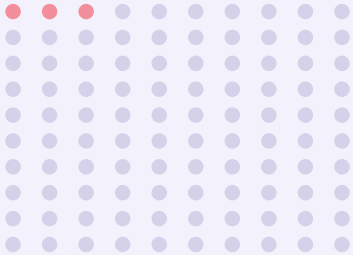
Multiple-choice percentages use all 103 respondents; options can overlap and totals can exceed 100%.

RISK PRESENTATION / MATERIAL FORMATS

Clearer risk. More ways to understand.

Q9 / PREFERRED RISK PRESENTATION

3 in 100



+ study size
+ follow-up duration

Illustrative wording, not an observed treatment outcome.

3 in 100 + study size + follow-up	**35.9%**	n=37
Combine different kinds of information	**22.3%**	n=23
Generally safe; some risks remain	**16.5%**	n=17
Not sure	**11.7%**	n=12
Low incidence; most can tolerate it	**9.7%**	n=10
No safety issues beyond expectations	**3.9%**	n=4

Q11 / PREFERRED MATERIAL FORMATS

51.5%
Illustrations / flowcharts
n=53

49.5%
One-page summary
n=51

45.6%
Short videos
n=47

Detailed expert explanations	**36.9% (38)**	Captions, audio and accessible versions	**16.5% (17)**
Real numbers / absolute-risk examples	**35.9% (37)**	Large print; less text	**13.6% (14)**
Frequently asked questions	**22.3% (23)**	Other	**1.0% (1)**
Questions with clinicians / researchers	**21.4% (22)**		

Q9 (single choice); Q11 (multiple choice) | n=103 | Source: ORCA aggregate survey results.
Multiple-choice percentages use all 103 respondents; options can overlap and totals can exceed 100%.

DESIGN PROPOSAL / RESEARCH BOUNDARIES

From the responses to a material structure

A proposed information hierarchy to pilot with the intended audience.

1 / DECISION SUMMARY

What might improve?
How large are the risks?
Who may be suitable?

2 / VISUAL EXPLANATION

Mechanism, research stage, measured outcomes and remaining uncertainty.

3 / OPTIONAL DETAIL & DISCUSSION

Study size, follow-up, alternatives, reliable sources, accessible versions and expert Q&A.

Q8

68.9%

prioritize actual improvement

Q10

49.5%

select unspecified improvement as a barrier

Q11

51.5%

prefer illustrations or flowcharts

NEXT STEP / TEST THE PROPOSAL

Recruit
Older adults and caregivers

→

Pilot
Use draft materials

→

Assess
Measure comprehension and accessibility

→

Revise
Improve the materials

Scope: 96 of 103 respondents (93.2%) are aged 18-30. Recruitment details and respondent-level records were not supplied. These aggregate results support descriptive priorities, not population estimates, individual response patterns, causal effects or evidence of material effectiveness.

Statistics: counts follow the supplied four-page survey summary; percentages use n=103 and are rounded to one decimal. Q6 retains the multiple-choice format in the source export. Full item wording appears in the questionnaire.